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REVISED GUIDELINES FOR THE IMPLEMENTATION OF
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THE MARINE ENVIRONMENT PROTECTION COMMITTEE,
RECALLING Article 38(c) of the Convention on the International Maritime Organization concerning the function of the Marine Environment Protection Cornnli t tee,
RECOGNIZING that Annex V of the International Convention for the Prevention from Ships, 1973, as modified by the Protocol of 1978 relating thereto (MARPOL 73/78) provides regulations for the prevention of pollution by garbage from ships,
RECOGNIZING ALSO the necessity of providing guidelines to assist Governments in developing and enacting domestic laws and regulations which give effect to and implement Annex V of MARPOL 73/78,
RECOGNIZING FURTHER that Part IV (Garbage) of the Guidelines on the Provision of Adequate Reception Facilities developed by the Organization in 1978,
BEING AWARE that the Committee, at its twenty-sixth session, modified the

RECOGNIZING FURTHER that Part IV (Garbage) of the Guidelines on the Provision of Adequate Reception Facilities developed by the Organization in 1978,
BEING AWARE that the Committee, at its twenty-sixth session, modified the above guidelines for garbage and developed guidelines for the implementation of Annex V of MARPOL 73/78 which were incorporated in the publication "MARPOL 73178, Consolidated Edition, 1991",
BEING ALSO AWARE that the Assembly at its seventeenth session adopted resolution A.719(17) on prevention of air pollution from ships, and requested the Committee and the Maritime Safety Committee to develop environmentally based standards for incineration of garbage and other ship-generated waste,
HAVING CONSIDERED the recommendations by the Sub-Committee on Ship Design and Equipment at its thirty-fifth session and the Sub-Committee on Bulk Chemicals at its twenty-second session regarding the standard specification for shipboard incinerators,
1. ADOPTS the Revised Guidelines for the Implementation of Annex V of MARPOL 73/78; and
2. RECOMMENDS the Governments to implement the provisions of Annex V of MAR POL 73/78 in accordance with the revised guidelines.

for shipboard incinerators,
1. ADOPTS the Revised Guidelines for the Implementation of Annex V of MARPOL 73/78; and
2. RECOMMENDS the Governments to implement the provisions of Annex V of MAR POL 73/78 in accordance with the revised guidelines.
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REVISED GUIDELINES* FOR THE IMPLEMENTATION OF ANNEX V OF MAR POL 73/78
The guidelines contained in MARPOL 73/78, Consolidated Edition, 1991, are amended as set out hereunder.
1 to replace paragraph 5.4.7 with the following:
"5.4.7 Shipboard incinerators should be designed, constructed, operated and maintained in accordance with the Standard Specification for Shipboard Incinerators set out in Appendix 2.'~
2 to amend the p"resent appendix as appendix 1.
3 to add the following as appendix 2:
Revised guidelines are to be prepared by the Secretariat incorporating the above amendments to the Guidelines contained in MARPOL 73178, Consolidated Edition, 1991.
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Revised guidelines are to be prepared by the Secretariat incorporating the above amendments to the Guidelines contained in MARPOL 73178, Consolidated Edition, 1991.
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Standard Specification for Shipboard Incinerators
Table of Contents
1 Scope
2 Definitions
3 Materials and manufacture
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Al Emission Standard for Shipboard Incinerators
A2 Fire Protection Requirements for Incinerators and Waste Stowage Spaces
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Standard specification for shipboard incinerators
1 Scope:
1.1 This specification covers the design, manufacture, performance, operation, functioning, and testing of incinerators intended to incinerate

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Standard specification for shipboard incinerators
1 Scope:
1.1 This specification covers the design, manufacture, performance, operation, functioning, and testing of incinerators intended to incinerate garbage and other shipboard wastes generated during the ship's normal service (i.e. maintenance, operational, domestic and cargo associated waste, excluding cargo associated wastes contaminated with Annex II and III substances as defined in the International Convention for the Prevention of Pollution From Ships, 1973, as modified by the Protocol of 1978 (MARPOL 73/78».
1.2 This specification applies to those incinerator plants with capacities up to 1,160 kW per unit.
1.3 This specification does not apply to systems on special incinerator ships, e.g., for burning industrial wastes such as chemicals, manufacturing residues, etc.
1.4 This specification does not address the electrical supply to the unit, nor the foundation connections and stack connections .
1.5 This specification provides emission requirements in annex AI, and fire protection requirements in annex A2 . Provisions for incinerators integrated

nor the foundation connections and stack connections .
1.5 This specification provides emission requirements in annex AI, and fire protection requirements in annex A2 . Provisions for incinerators integrated with heat recovery units and provisions for flue gas temperature are given in annex A3 and annex A4, respectively.
1.6 This standard may involve hazardous materials, operations, and equipment. This standard does not purport to address all of the safety problems associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use, including possible port State limitations.
2 Definitions
2.1 Ship means a vessel of any type whatsoever operating in the marine environment and includes hydrofoil boats, air-cushioned vehicles, submersibles, floating craft and fixed or floating platforms.
2.2 Incinerator means shipboard facilities for incinerating solid wastes approximating in composition to household waste and liquid wastes arising from the operation of the ship, e.g., domestic waste, cargo-associated waste,

2.2 Incinerator means shipboard facilities for incinerating solid wastes approximating in composition to household waste and liquid wastes arising from the operation of the ship, e.g., domestic waste, cargo-associated waste, maintenance waste, operational waste, cargo residues, and fishing gear, etc. These facilities may be designed to use or not to use the heat energy produced.
2.3 Garbage means all kinds of victual, domestic and operational waste excluding fresh fish and parts thereof, generated during normal operation of the ship as defined in Annex V to MAR POL 73/78.
2.4 Waste means useless, unneeded or superfluous matter which is to be discarded.
2.5 Food wastes are any spoiled or unspoiled victual substances, such as fruits, vegetables, dairy products, poultry, meat products, food scraps, food particles, and all other materials contaminated by such wastes, generated aboard ship, principally in the galley and dining areas.
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2.6 Plastic means a solid material which contains as an essential ingredient one or more synthetic organic high polymers and which is formed (shaped) during either manufacture of the polymer or the fabrication into a finished product by heat and/or pressure. Plastics have material properties ranging from hard and brittle to soft and elastic~ Plastics are used for a variety of marine purposes including, but not limited to, packaging (vapour-proof barriers, bottles, containers, liners), ship construction (fibreglass and laminated structures, siding, piping, insulation, flooring, carpets, fabrics, paints and finishes, adhesives, electrical and electronic components), disposable eating utensils and cups, bags, sheeting, floats, fishing nets, strapping bands, rope and line.
2.7 Domestic waste means all types of food wastes, sewage and wastes generated in the living spaces on board the ship.

disposable eating utensils and cups, bags, sheeting, floats, fishing nets, strapping bands, rope and line.

2.7 Domestic waste means all types of food wastes, sewage and wastes generated in the living spaces on board the ship.

2.8 Cargo-associated waste means all materials which have become wastes as a result of use on board a ship for cargo stowage and handling. Cargo-associated waste includes but is not limited to dunnage, shoring pallets, lining and packing materials, plywood, paper, cardboard, wire, and steel strapping.

2.9 Maintenance waste means materials collected by the engine department and the deck department while maintaining and operating the vessel, such as soot, machinery deposits, scraped paint, deck sweeping, wiping wastes, oily rags, etc.

2.10 Operational wastes means all cargo-associated wastes and maintenance waste (including ash and clinkers), and cargo residues defined as garbage in 2.13.

2.11 Sludge oil means sludge from fuel and lubricating oil separators, waste lubricating oil from main and auxiliary machinery, waste oil from bilge water separators, drip trays, etc.

2.13.
2.11 Sludge oil means sludge from fuel and lubricating oil separators, waste lubricating oil from main and auxiliary machinery, waste oil from bilge water separators, drip trays, etc.
2.12 Oily rags are rags which have been saturated with oil as controlled in Annex I to the Convention. Contaminated rags are rags which have been saturated with a substance defined as a harmful substance in the other Annexes to MARPOL 73/78.
2.13 Cargo residues for the purposes of this standard are defined as the remnants of any cargo material on board that cannot be placed in proper cargo holds (loading excess and spillage) or which remains in cargo holds and elsewhere after unloading procedures are completed (unloading residual and spillage). However, cargo residues are expected to be in small quantities.
2.14 Fishing gear is defined as any physical device or part thereof or combination of items that may be placed on or in the water with the intended purpose of capturing, or controlling for subsequent capture, living marine or freshwater organisms.
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purpose of capturing, or controlling for subsequent capture, living marine or freshwater organisms.
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3 Materials and manufacture:
3.1 The materials used in the individual parts of the incinerator are to be suitable for the intended application with respect to heat resistance, mechanical properties, oxidation, corrosion, etc., as in other auxiliary marine equipment.
3.2 Piping for fuel and sludge oil should be seamless steel of adequate strength and to the satisfaction of the Administration. Short lengths of steel, or annealed copper nickle, nickle copper, or copper pipe and tubing may be used at the burners. The use of nonmetallic materials for fuel lines is prohibited. Valves and fittings may be threaded in sizes up to and including 60 mm 0.0. (outside diameter), but threaded unions are not to be used on pressure lines in sizes 33 mm 0.0. (outside diameter) and over.
3.3 All rotating or moving mechanical and exposed electrical parts should be protected against accidental contact.

pressure lines in sizes 33 mm 0.0. (outside diameter) and over.
3.3 All rotating or moving mechanical and exposed electrical parts should be protected against accidental contact.
3.4 Incinerator walls are to be protected with insulated fire bricks/refractory and a cooling system. Outside surface temperature of the incinerator casing being touched during normal operations should not exceed 20°C above ambient temperature.
3.5 Refractory should be resistant to thermal shocks and resistant to normal ship's vibration. The refractory design temperature should be equal to the combustion chamber design temperature plus 20~. (See 4.1)
3.6 Incinerating systems should be designed such that corrosion will be minimized on the inside of the systems.
3.7 In systems equipped for incinerating liquid wastes, safe ignition and maintenance of combustion must be ensured, e.g., by a supplementary burner.
3.8 The combustion chamber(s) should be designed for easy maintenance of all internal parts including the refractory and insulation.
3.9 The combustion process should take place under negative pressure which

3.8 The combustion chamber(s) should be designed for easy maintenance of all internal parts including the refractory and insulation.
3.9 The combustion process should take place under negative pressure which means that the pressure in the furnace under all circumstances should be lower than the ambient pressure in the room where the incinerator is installed. A flue gas fan may be fitted to secure negative pressure.
3.10 The incinerating furnace may be charged with solid waste either by hand or automatically. In every case, fire dangers should be avoided and charging should be possible without danger to the operating personnel.
For instance, where charging is carried out by hand, a charging lock may be provided which ensures that the charging space is isolated from the fire box as long as the filling hatch is open.
Where charging is not effected through a charging lock, an interlock should be installed to prevent the charging door from opening while the incinerator is in operation with burning of garbage in progress or while the furnace temperature is above 220°C.
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incinerator is in operation with burning of garbage in progress or while the furnace temperature is above 220°C.
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3.11 Incinerators equipped with a feeding sluice or system should ensure that the material charged will move to the combustion chamber. Such system should be designed such that both operator and environment are protected from hazardous exposu~e.
3.12 Interlocks should be installed to prevent ash removal doors from opening while burning is in progress or while the furnace temperature is above 220°C.
3.13 The incinerator should be provided with a safe observation port of the combustion chamber in order to provide visual control of the burning process and waste accumulation in the combustion chamber. Neither heat, flame, nor particles should be able to pass through the observation port. An example of a safe observation port is high-temperature glass with a metal closure.
3.14 Electrical requirements
3.14.1 International Electrotechnical Commission (IEC) Standards,

particles should be able to pass through the observation port. An example of a safe observation port is high-temperature glass with a metal closure.

3.14 Electrical requirements

3.14.1 International Electrotechnical Commission (IEC) Standards, particularly IEC Publication 92 - Electrical Installations in Ships and Mobile and Fixed Offshore Units, are applicable for this equipment.

3.14.2 Electrical installation requirements should apply to all electrical equipment, including controls, safety devices, cables, and burners and incinerators.

3.14.2.1 A disconnecting means capable of being locked in the open position should be installed at an accessible location at the incinerator so that the incinerator can be disconnected from all sources of potential. This disconnecting means should be an integral part of the incinerator or adjacent to it. (See 5.1)

3.14.2.2 All uninsulated live metal parts should be guarded to avoid accidental contact.

3.14.2.3 The electrical equipment should be so arranged so that failure of this equipment will cause the fuel supply to be shut off.

3.14.2.4 All electrical contacts of every safety device installed in the

accidental contact.
3.14.2.3 The electrical equipment should be so arranged so that failure of this equipment will cause the fuel supply to be shut off.
3.14.2.4 All electrical contacts of every safety device installed in the control circuit should be electrically connected in series. However, special consideration should be given to arrangements when certain devices are wired in parallel.
3.14.2.5 All electrical components and devices should have a voltage rating commensurate with the supply voltage of the control system.
3.14.2.6 All electrical devices and electric equipment exposed to the weather should be according to IEC Publication 92.201, Table V.
3.14.2.7 All electrical and mechanical control devices should be of a type tested and accepted by a nationally recognized testing agency, according to international standards.
3 . 14.2.8 The design of the control circuits should be such that limit and primary safety controls should directly open a circuit that functions to interrupt the supply of fuel to combustion units.
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primary safety controls should directly open a circuit that functions to interrupt the supply of fuel to combustion units.
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3.14.3 Overcurrent protection
3.14.3.1 Conductors for interconnecting wiring that is smaller than the supply conductors should be provided with overcurrent protection based on the size of the smallest interconnecting conductors external to any control box, according to IEC's rules.
3.14.3.2 Overcurrent protection for interconnecting wiring should be located at the point where the smaller conductors connect to the larger conductors. However, overall overcurrent protection is acceptable if it is sized on the basis of the smallest conductors of the interconnecting wiring, or according to IEC's requirements.
3.14.3.3 Overcurrent protection devices should be accessible and their function should be identified.
3.14.4 Motors
3.14.4.1 All electric motors should have enclosures corresponding to the environment where they are located, at least IP 44, according to

function should be identified.

3.14.4 Motors

3.14.4.1 All electric motors should have enclosures corresponding to the environment where they are located, at least IP 44, according to IEC Publication 529.

3.14.4.2 Motors should be provided with a corrosion-resistant nameplate specifying information in accordance with IEC Publication 92-301.

3.14.4.3 Motors should be provided with running protection by means of integral thermal protection, by overcurrent devices, or a combination of both in accordance with manufacturer's instruction that should be in accordance with IEC Publication 92-202.

3.14.4.4. Motors should be rated for continuous duty and should be designed for an ambient temperature of 45°C or higher.

3.14.4.5 All motors should be provided with terminal leads or terminal screws in terminal boxes integral with, or secured to, the motor frames.

3.14.5 Ignition system

3.14.5.1 When automatic electric ignition is provided, it should be accomplished by means of either a high-voltage electric spark, a high-energy electric spark, or a glow coil.

3.14.5.2 Ignition transformers should have an enclosure corresponding to the

accomplished by means of either a high-voltage electric spark, a high-energy electric spark, or a glow coil.
3.14.5.2 Ignition transformers should have an enclosure corresponding to the environment where they are located, at least IP 44 according to IEC Publication 529.
3.14.5.3 Ignition cable should conform to the requirements of IEC Publication 92-503.
3.14.6 Wiring
3.14.6.1 All wiring for incinerators should be rated and selected in accordance with IEC Publication 92-352.
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3.14.7.1 Means should be provided for grounding the major metallic frame or assembly of the incinerators.
3.14.7.2 Noncurrent carrying enclosures, frames and similar parts of all electrical components and devices should be bonded to the main frame or assembly of the incinerator. Electrical components that are bonded by their installation do not require a separate bonding conductor.
3.14.7.3 When an insulated conductor is used to bond electrical components

assembly of the incinerator. Electrical components that are bonded by their installation do not require a separate bonding conductor.

3.14.7.3 When an insulated conductor is used to bond electrical components and devices, it should show a continuous green colour, with or without a yellow stripe.

4 Operating requirements:

4.1 The incinerator system should be designed and constructed for operation with the following conditions:

Maximum Gombustion chamber
flue gas outlet temperature

Minimum combustion chamber
flue gas outlet temperature

Preheat temperature of
combustion chamber

For Batch Loaded Incinerators, there are no preheating requirements. However, the incinerator should be designed that the temperature in the actual combustion space should reach 600 0 e within 5 minutes after start.

Prepurge, before ignition:

Time between restarts:

Postpurge, after shut-off
fuel oil:

Incinerator discharge gases:

at least 4 air changes in the
chamber(s) and stack, but not
less than 15 seconds.

at least 4 air changes in the
chamber(s) and stack, but not
less than 15 seconds.

not less than 15 seconds after
the closing of the fuel oil
valve.

Minimum 6~ 02'

chamber(s) and stack, but not
less than 15 seconds.
at least 4 air changes in the
chamber(s) and stack, but not
less than 15 seconds.
not less than 15 seconds after
the closing of the fuel oil
valve.
Minimum 6~ 02'
4.2 Outside surface of combustion chamber(s) should be shielded from contact such that people in normal work situations will not be exposed to extreme heat (20 0 e above ambient temperature) or direct contact of surface temperatures exceeding 60 o e. Examples for alternatives to accomplish this are a double jacket with an air flow in between or an expanded metal jacket.
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4.3 Incinerating systems are to be operated with underpressure (negative pressure) in the combustion chamber such that no gases or smoke can leak out to the surrounding areas.
4.4 The incinerator should have warning plates attached in a prominent location on the unit, warning against unauthorized opening of doors to combustion chamber(s) during operation and against overloading the incinerator with garbage.

4.4 The incinerator should have warning plates attached in a prominent location on the unit, warning against unauthorized opening of doors to combustion chamber(s) during operation and against overloading the incinerator with garbage.
4.5 The incinerator should have instruction plate(s) attached in a prominent location on the unit that clearly addresses the following:
4.5.1 Cleaning ashes and slag from the combustion chamber(s) and cleaning of combustion air openings before starting the incinerator (where applicable).
4.5.2 Operating procedures and instructions. These should include proper start-up procedures, normal shut-down procedures, emergency shut-down procedures, and procedures for loading garbage (where applicable).
4.6 To avoid building up of dioxins, the flue gas should be shock-cooled to a maximum 350°C right after the incinerator.
5 Operating controls
5.1 The entire unit should be capable of being disconnected from all sources of electricity by means of one disconnect switch located near the incinerator. (See 3.14.2.1)
5.2 There should be an emergency stop switch located outside the compartment

sources of electricity by means of one disconnect switch located near the incinerator. (See 3.14.2.1)

5.2 There should be an emergency stop switch located outside the compartment which stops all power to the equipment. The emergency stop switch should also be able to stop all power to the fuel pumps. If the incinerator is equipped with a flue gas fan, the fan should be capable of being restarted independently of the other equipment on the incinerator.

5.3 The control equipment should be so designed that any failure of the following equipment will prevent continued operations and cause the fuel supply to be cut off.

5.3.1 Safety thermostat/draft failure

5.3.1.1 A flue gas temperature controller, with a sensor placed in the flue gas duct, should be provided that will shut down the burner if the flue gas temperature exceeds the temperature set by the manufacturer for the specific design.

5.3.1.2 A combustion temperature controller, with a sensor placed in the combustion chamber, should be provided that will shut down the burner if the combustion chamber temperature exceeds the maximum temperature.

design.
5.3.1.2 A combustion temperature controller, with a sensor placed in the combustion chamber, should be provided that will shut down the burner if the combustion chamber temperature exceeds the maximum temperature.
5.3.1.3 A negative pressure switch should be provided to monitor the draft and the negative pressure in the combustion chamber. The purpose of this negative pressure switch is to ensure that there is sufficient draft/negative pressure in the incinerator during operations. The circuit to the program relay for the burner will be opened and an alarm activated before the negative pressure rises to atmospheric pressure.
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5.3.2.1 The incinerator should have a flame safeguard control consisting of a flame ~ensing element and assoc i ated equipment for shut down of the unit in the event of ignition failure a nd fl'~e 'fi:tilure'duringthe firing cycle. The flame safeguard control should be so designed that the failure of any

flame ~ensing element and assoc i ated equipment for shut down of the unit in the event of ignition failure a nd fl'~e 'fi:tilure'duringthe firing cycle. The flame safeguard control should be so designed that the failure of any component will cause a safety shut down.

5.3.2.2 The flame safeguard control should be capable of closing the fuel valves in not more than 4 seconds after a flame failure.

5.3.2.3 The flame safeguard control should provide a trial-for-ignition period of not more that 10 seconds during which fuel may be supplied to establish flame. If flame is not established within 10 seconds, the fuel supply to the burners should be immediately shut off automatically.

5.3.2.4 Whenever the flame safeguard control has operated because of failure of ignition, flame failure, or failure of any component, only one automatic restart may be provided. If this is not successful then manual reset of the flame safeguard control should be required for restart.

5.3.2.5 Flame safeguard controls of the thermostatic type, such as stack switches and pyrostats operated by means of an open bimetallic helix, are prohibited.

flame safeguard control should be required for restart.
5.3.2.5 Flame safeguard controls of the thermostatic type, such as stack switches and pyrostats operated by means of an open bimetallic helix, are prohibited.
5.3.2.6 If fuel oil pressure drops below that set by. the manufacturer, a failure and lock out of the program relay should result. This also applies to sludge oil used as a fuel. (Applies where pressure is important for the combustion process or a pump is not an integral part of the burner.)
5.3.3 Loss of power
If there is a loss of power to the incinerator control/alarm panel (not remote alarm panel), the system should shut down.
5.4 Fuel supply
Two fuel control solenoid valves should be provided in series in the fuel supply line to each burner. On multiple burner units, a valve on the main fuel supply line and a valve at each burner will satisfy this requirement. The valves should be connected electrically in parallel so that both operate simultaneously.
5.5 Alarms
5.5.1 An outlet for an audible alarm should be provided for connection to a local alarm system or a central alarm system. When a failure occurs, a

simultaneously.
5.5 Alarms
5.5.1 An outlet for an audible alarm should be provided for connection to a local alarm system or a central alarm system. When a failure occurs, a visible indicator should show what caused the failure. (The indicator may cover more than one fault condition.)
5.5.2 The visible indicators should be designed so that, where failure is a safety related shutdown, manual reset is required.
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5.6 After shutdown of the oil burner, prov~s~on should be made for the fire box to cool sufficiently. (As an example, of how this may be accomplished, the exhaust fan or ejector could be designed to continue to operate. This would not apply in the case of an emergency manual trip.)
6 Other requirements
6 . 1 Documentation
A complete instruction and maintenance manual with drawings, electric diagrams, spare parts list, etc., should be furnished with each incinerator.
6.2 Installation
All devices and components should, as fitted in the ship, be designed to

A complete instruction and maintenance manual with drawings, electric diagrams, spare parts list, etc., should be furnished with each incinerator.

6.2 Installation

All devices and components should, as fitted in the ship, be designed to operate when the ship is upright and when inclined at any angle of list up to and including 15° either way under static conditions and 22.5° under dynamic conditions (rolling) either way and simultaneously inclined dynamically (pitching) 7.5° by bow or stern.

6.3 Incinerator

6.3.1 Incinerators are to be fitted with an energy source with sufficient energy to ensure a safe ignition and complete combustion. The combustion is to take place at sufficient negative pressure in the combustion chamber(s) to ensure no gases or smoke leaking out to the surrounding areas. (See 5.3.1.3)

6.3.2 A driptray is to be fitted under each burner and under any pumps, strainers, etc., that require occasional examination.

7 Tests

7 . 1 Prototype tests

An operating test for the prototype of each design should be conducted, with a test report completed indicating results of all tests. The tests should be

strainers, etc., that require occasional examination.

7 Tests

7 . 1 Prototype tests

An operating test for the prototype of each design should be conducted, with a test report completed indicating results of all tests. The tests should be conducted to ensure that all of the control components have been properly installed and that all parts of the incinerator, including controls and safety devices, are in satisfactory operating condition. Tests should include those described in section 7.3 below.

7.2 Factory tests

For each unit, if preas sembled, an operating test should be conducted to ensure that all of the control components have been properly installed and that all parts of the incinerator, including controls and safety devices, are in satisfactory operating condition. Tests should include those described in 7.3 below.

7.3 Installation tests

An operating test after installation should be conducted to ensure that all of the control components have been properly installed and . that all parts of the incinerator, including controls and safety devices, are in satisfactory operating condition.

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of the incinerator, including controls and safety devices, are in satisfactory operating condition.
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7.3.1 Flame safeguard. The operation of the flame safeguard system should be verified by causing flame and ignition failures. Operation of the audible alarm (where applicable) and visible indicator should be verified. The shutdown times should be verified.
7.3.2 Limit controls. Shutdown due to the operation of the limit controls should be verified.
7.3.2.1 Oil pressure limit control. The lowering of the fuel oil pressure below the value required for safe combustion should initiate a safety shutdown.
7.3.2.2 Other interlocks. Other interlocks provided should be tested for proper operation as specified by the unit manufacturer.
7.3.3 Combustion controls. The combustion controls should be stable and operate smoothly.
7.3.4 Programming controls. Programming controls should be verified as controlling and cycling the unit in the intended manner. Proper prepurge,

7.3.3 Combustion controls. The combustion controls should be stable and operate smoothly.
7.3.4 Programming controls. Programming controls should be verified as controlling and cycling the unit in the intended manner. Proper prepurge, ignition, postpurge, and modulation should be verified. A stopwatch should be used for verifying intervals of time.
7.3.5 Fuel supply controls. The satisfactory operation of the two fuel control solenoid valves for all conditions of operation and shutdown should be verified.
7.3.6. Low voltage test. A low voltage test should be conducted to satisfactorily demonstrate that the fuel supply to the burners will be automatically shut off before an incinerator malfunction results from the reduced voltage.
7.3.7 Switches. All switches should be tested to verify proper operation.
8 Certification
8.1 Manufacturer's certification that an incinerator has been constructed in accordance with this standard should be provided (by letter, certificate, or in the instruction manual).
9 Marking
9.1 Each incinerator should be permanently marked indicating:
9.1.1 Manufacturer's name or trademark.

accordance with this standard should be provided (by letter, certificate, or in the instruction manual).

9 Marking

9.1 Each incinerator should be permanently marked indicating:

9.1.1 Manufacturer's name or trademark.

9.1.2 Style, type, model or other manufacturer's designation for the incinerator.

9.1.3 Capacity - to be indicated by net designed heat release of the incinerator in heat units per timed period; for example, British Thermal Units per hour, megajoules per hour, kilocalories per hour.

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10 Quality assurance

10.1 Incinerators should be designed, manufactured and tested in a manner that ensures they meet the requirements of this standard.

10.2 The incinerator manufacturer should have a quality system that meets ISO 9001, "Quality Systems - Model for Quality Assurance in Design/Development, Production, Installation and Servicing". The quality system should consist of elements necessary to ensure that the production

meets ISO 9001, "Quality Systems - Model for Quality Assurance in Design/Development, Production, Installation and Servicing". The quality system should consist of elements necessary to ensure that the production quality of the incinerators are designed, tested and marked in accordance with this standard. At no time should an incinerator be sold with this standard designation that does not meet the requirements herein (see Certification).

Al - EMISSION STANDARD FOR SHIPBOARD INCINERATORS WITH CAPACITIES OF UP TO 1,160 kW

Minimum information to be provided

A1.1 An IMO TYPE APPROVAL CERTIFICATE should be required for each shipboard incinerator. In order to obtain such certificate, the incinerator should be designed and built to an IMO approved standard. Each model should go through a specified type approval test operation at the factory or an approved test facility, and under the responsibility of the Administration.

A1.2 TYPE APPROVAL TEST SHOULD INCLUDE MEASURING OF THE FOLLOWING PARAMETERS:

Max capacity

Pilot fuel consumption

02 Average

in combustion chamber/zone

CO Average in flue gas

Soot number average

Combustion chamber flue gas

A1.2 TYPE APPROVAL TEST SHOULD INCLUDE MEASURING OF THE
FOLLOWING PARAMETERS:
Max capacity
Pilot fuel consumption
02 Average
in combustion chamber/zone
CO Average in flue gas
Soot number average
Combustion chamber flue gas
outlet temperature average:
Amount of unburned
components in ashes
Al.3 DURATION OF TEST OPERATION
For sludge oil burning
For solid waste burning
W/9096D/ljt/EWP
kW or kcal/h
kg/h of specified waste
kg/h per burner
kg/h per burner
mg/MJ
Bacharach or ringelman
Scale
~ by weight
6-8 hours
6-8 hours
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Al.4 FUEL/WASTE SPECIFICATION FOR TYPE APPROVAL TEST (~ BY WEIGHT)
Sludge oil consisting of:
Solid waste (class 2)
consisting of:
Classes of waste
75% sludge oil from heavy fuel oil
5% waste lubricating oil
20% emulsified water
50% Food Waste
50% rubbish Containing
Approx. 30% paper,
" 40~ Cardboard,
"
"
10~ Rags,
20~ Plastic
The mixture will have up to 50%
moisture and 7% incombustible
solids
Reference: Waste Classification from Incinerator Institute of America.

50% rubbish Containing
Approx. 30% paper,
" 40~ Cardboard,
"
"
10~ Rags,
20~ Plastic
The mixture will have up to 50%
moisture and 7% incombustible
solids
Reference: Waste Classification from Incinerator Institute of America.
Class 0 Trash, a mixture of highly combustible waste such as paper, cardboard, wood boxes, and combustible floor sweepings, with up to 10% by weight of plastic bags, coated paper, laminated paper, treated corrugated cardboard, oil rags, and plastic or rubber scraps. This type of waste contains up to 10% moisture, 5% incombustible solids and has a heating value of about 19,700 kJ/kg as fired.
Class 1 Rubbish, a mixture of combustible waste such as paper, cardboard cartons, wood scrap, foliage, and combustible floor sweepings. The mixture contains up to 20% by weight of galley or cafeteria waste, but contains little or no treated papers, plastic, or rubber wastes. This type of waste contains 25% moisture, 10% incombustible solids and has a heating value of about 15,100 kJ/kg as fired.
Class 2 Refuse, consisting of approximately even mixture of rubbish and garbage by weight. This type waste is common to passenger ships

solids and has a heating value of about 15,100 kJ/kg as fired.
Class 2 Refuse, consisting of approximately even mixture of rubbish and garbage by weight. This type waste is common to passenger ships occupancy, consisting of up to 50% moisture, 7% incombustible solids and has a heating value of about 10,000 kJ/kg as fired.
Class 3 Garbage, consisting of animal and vegetable wastes from restaurants, cafeterias, galleys, sick bays and like installations. This type of waste contains up to 70% moisture, up to 5% incombustible solids and has a heating value range of about 2,300 kJ/kg as fired.
Class 4 Aquatic life forms and animal remains, consisting of carcasses, organs and solid organic wastes from vessels carrying animal type cargos, consisting of up to 85% moisture, 5% incombustible solids and having a heating value range of about 2,300 kJ/kg as fired.
Class 5 By-product waste, liquid or semi-liquid, such as tar, paints, solvents, sludge, oil, waste oil, etc., from shipboard operations.
BTU values must be determined by the individual materials to be destroyed.
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solvents, sludge, oil, waste oil, etc., from shipboard operations.
BTU values must be determined by the individual materials to be
destroyed.
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Class 6 Solid by-product waste, such as rubber, plastics, wood waste, etc.,
from industrial operations. BTU values must be determined by the
individual materials to be destroyed.
Calorific values kJ/Kg kcal/kg
Vegetable and putrescibles 5,700 1,360
Paper 14,300 3,415
Rag 15,500 3,700
Plastics 36,000 8,600
Oil sludge 36,000 8,600
Sewage sludge 3,000 716
Densities kg/m 3
Paper (loose) 50
Refuse (75% wet) 720
Dry rubbish 110
Scrap wood 190
Wood sawdust 220
Density of loose general waste generated on board ship will be about
130 kg/m 3
Al.5 REQUIRED EMISSION STANDARDS TO BE VERIFIED BY TYPE APPROVAL TEST
02 in combustion chamber 6 - 12%
CO in flue gas maximum average 200 mg/MJ
Soot number maximum average BACHARACH 3 or RINGELMAN 1
Unburned components in
ash residues
Combustion chamber flue gas
outlet temperature range

02 in combustion chamber 6 - 12%
CO in flue gas maximum average 200 mg/MJ
Soot number maximum average BACHARACH 3 or RINGELMAN 1
Unburned components in
ash residues
Combustion chamber flue gas
outlet temperature range
(A higher soot number is acceptable
only during very short periods such
as starting up)
Max 10% by Weight
900 - 1,200°C
A high temperature in the actual combustion chamber/zone is an absolute requirement in order to obtain a complete and smoke free incineration, including that of plastic and other synthetic materials while minimizing DIOXINE, VOC (Volatile Organic Compounds), and emissions.
Al.6 FUEL RELATED EMISSION
Al.6.1 Even with good incineration technology the emission from an incinerator will depend on the type of material being incinerated. If for instance a vessel has bunkered a fuel with high sulphur content, then sludge oil from separators which is burned in the incinerator will lead to emission of Sox. But again, the Sox emission from the incinerator would only amount to less than one per cent of the Sox discharged with the exhaust from main and auxiliary engines.
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of Sox. But again, the Sox emission from the incinerator would only amount to less than one per cent of the Sox discharged with the exhaust from main and auxiliary engines.
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A1.6.2 Principal organic constituents (POC) cannot be measured on a continuous basis. Specifically, there are no instruments with provision for continuous time telemetry that measures POC, HCl, or waste destruction efficiency, to gate. These measuremepts can only be made using grab sample
'~ '. ' , :- 'l' .' J', t- :. " ' "
approaches where the sample is returned to a laboratory for analys~s. In the case of organic constituents (undestroyed wastes), the laboratory work requires considerable time to complete. Thus, continuous emission control can only be assured by secondary measurements.
A1.6.3 ON-BOARD OPERATION/EMISSION CONTROL
For a shipboard incinerator with IMO TYPE APPROVAL, emission control/monitoring should be limited to the following:
.1 Control/monitor 02 content in combustion chamber (spot checks only) .

A1.6.3 ON-BOARD OPERATION/EMISSION CONTROL
For a shipboard incinerator with IMO TYPE APPROVAL, emission control/monitoring should be limited to the following:
.1 Control/monitor 02 content in combustion chamber (spot checks only) .
. 2 Control/monitor temperature in combustion chamber flue gas outlet.
By continuous (auto) control of the incineration process, ensure that the above mentioned two parameters are kept within the prescribed limits. This mode of operation will ensure that particulates and ash residue contain only traces of organic constituents.
A1.7 PASSENGER/CRUISE SHIPS WITH INCINERATOR INSTALLATIONS HAVING A TOTAL CAPACITY OF MORE THAN 1,160 kW
A1.7.l On board this type of vessel, the following conditions will probably exist:
.1 Generation of huge amounts of burnable waste with a high content of plastic and synthetic materials .
2 Incinerating plant with a high capacity operating continuously over long periods
3 This type of vessel will often be operating in very sensitive coastal areas.
A1.7.2 In view of the fuel related emission from a plant with such a high capacity , installation of a flue gas sea water scrubber should be

long periods
3 This type of vessel will often be operating in very sensitive
coastal areas.
A1.7.2 In view of the fuel related emission from a plant with such a high
capacity , installation of a flue gas sea water scrubber should be
considered. This installation can perform an efficient after-cleaning of the
flue gases, thus minimizing the content of:
HC1
SOX
PARTICULATE MATTER
A1.7.3 Any restriction on NITROGEN OXIDE (NOX) should only be considered in
connection with possible future regulations on pollution from the vessel's
total pollution, i.e., main and auxiliary machinery, boilers, etc.
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A2 - FIRE PROTECTION REQUIREMENTS FOR INCINERATORS AND WASTE STOWAGE SPACES
For the purpose of construction, arrangement and insulation, incinerator
spaces and waste stowag e spaces should be treated as category A machinery
spaces (SOLAS II-2/3.19) and service spaces, (SOLAS 11-2/3.12), respectively.
To minimize the fire hazards these spaces represent, the following SOLAS

spaces and waste stowag e spaces should be treated as category A machinery spaces (SOLAS II-2/3.19) and service spaces, (SOLAS 11-2/3.12), respectively. To minimize the fire hazards these spaces represent, the following SOLAS requirements in chapter 11-2 should be applied:

A2.1 For passenger vessels carrying more than 36 passenger:

.1 regulation 26.2.2(12) should apply to incinerator and combined incinerator/waste storage spaces, and the flue uptakes from such spaces; and

. 2 regulation 26.2.2(13) should apply to waste storage spaces and garbage chutes connected thereto.

A2.2 For all other vessels including passenger vessels carrying not more than 36 passengers:

. 1 regulation 44.2.2(6) should apply to incinerator and combined incinerator/waste spaces, and the flue uptakes from such spaces; and

.2 regulation 44.2.2(9) should apply to waste storage spaces and garbage chutes connected thereto.

A2.3 Incinerators and waste stowage spaces located on weather decks (regulation 11-2/3.(17» need not meet the above requirements but should be located:

.1 as far aft on the vessel as possible,

. 2 not less than 3 m from entrances, air inlets and openings

(regulation 11-2/3.(17» need not meet the above requirements but should be located:

.1 as far aft on the vessel as possible,

. 2 not less than 3 m from entrances, air inlets and openings to accommodations, service spaces and control stations,

. 3 not less than 5 m measured horizontally from the nearest hazardous area, or vent outlet from a hazardous area; and

.4 not less than 2 m should separate the incinerator and the waste material storage area, unless physically separated by a structural fire barrier.

A2.4 A fixed fire detection and fire-extinguishing system should be installed in enclosed spaces containing incinerators, in combined incinerator/waste storage spaces, and in any waste storage space in accordance with the following table:

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Incinerator space

Waste storage space

Automatic sprinkler system

X

X

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Fixed fire extinguishing system

X

Fixed fire detection system

X

waste storage space
Incinerator space
Waste storage space
Automatic
sprinkler
system
X
X
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Fixed fire
extinguishing
system
X
Fixed fire
detection
system
X
A2 . 5 Where an incinerator or waste storage space is located on weather decks it must be accessible with two means of fire extinguishment; either fire hoses, semi-portable fire extinguishers, fire monitors or combination of any two of these extinguishing devices. A fixed fire-extinguishing system is acceptable as one means of extinguishment.
A2.6 Flue uptake piping/ducting should be led independently to an appropriate terminus via a continuous funnel or trunk.
A3 - INCINERATORS INTEGRATED WITH HEAT RECOVERY UNITS
A3.l The flue gas system, for incinerators where the flue gas is led through a heat recovery device, should be designed so that the incinerator can continue operation with the economizer coils dry. This may be accomplished with bypass dampers if needed.
A3.2 The incinerator unit should be equipped with a visual and an audible alarm in case of loss of feed-water.
A3.3 The gas-side of the heat recovery device should have equipment for

with bypass dampers if needed.
A3.2 The incinerator unit should be equipped with a visual and an audible alarm in case of loss of feed-water.
A3.3 The gas-side of the heat recovery device should have equipment for proper cleaning. Sufficient access should be provided for adequate inspection of external heating surfaces.
A4 - FLUE GAS TEMPERATURE
A4.1 When deciding upon the type of incinerator, consideration should be given as to what the flue gas temperature will be. The flue gas temperature can be a determining factor in the selection of materials for fabricating the stack. Special high temperature material may be required for use in fabricating the stack when the flue gas temperatures exceed 430°C.
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ANNEX
FORM OF IMO TYPE APPROVAL CERTIFICATE FOR
SHIPBOARD INCINERATORS WITH CAPACITIES OF UP TO 1,160 KW
CERTIFICATE OF SHIPBOARD INCINERATOR
NAME OF ADMINISTRATION
BADGE
OR
CYPHER
This is to certify that the shipboard incinera,tor listed has been

FORM OF IMO TYPE APPROVAL CERTIFICATE FOR
SHIPBOARD INCINERATORS WITH CAPACITIES OF UP TO 1,160 KW
CERTIFICATE OF SHIPBOARD INCINERATOR
NAME OF ADMINISTRATION
BADGE
OR
CYPHER
This is to certify that the shipboard incinera,tor listed has been
examined and tested in accordance with the requirement of the standard for
shipboard incinerators for disposing of ship-generated waste appended to the
Guidelines for the Implementation of Annex V of MARPOL 73/78.
Incinerator manufactured by .....
Style, type or model for the incinerator*
Max. capacity
O2 Average
in combustion chamber/zone
CO Average in flue gas
Soot number average
Combustion chamber flue gas
outlet temperature average
Amount of unburned components
in ashes
kW or kcal/h
kg/h of specified waste
kg/h per burner
'It,
mg/MJ
Bacharach or ringelman scale
'It, by weight
A copy of this certificate should be carried on board a yessel fitted
with this equipment at all times.
Signed ..... .
Official stamp Administration of ... .
Dated this .. day of .... , ... 19 ..
* Delete as appropriate.
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with this equipment at all times.
Signed ..... .
Official stamp Administration of ... .
Dated this .. day of .... , ... 19 ..
* Delete as appropriate.
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